

# Reese Ludwick

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## Summary

Forward-deployed AI engineer who builds and *runs* agent systems 0→1 in production — not prototypes that die in a notebook. Shipped MCP servers and LangGraph multi-agent workflows for enterprise R&D, plus two production-complexity agent platforms. Embeds with stakeholders to deliver working automation under ambiguity.

## Experience

**Freelance AI Engineer & Full-Stack Developer — Independent Contract (WorldCovers / APMC), Remote — May 2026 – Present**

- Embedded with stakeholders to ship features across production web properties (chinaoverprints.com, ianthom.org) and WorldCovers, a philatelic cataloging platform.
- Deliver workflow automation against a 14-table PostgreSQL schema powering a Django REST backend with a React / TypeScript frontend.
- Scope, build, and deploy production work on compressed timelines with AI-augmented engineering (Claude Code) — forward-deployed delivery under ambiguity.

**Software Associate — Cognizant Technology Solutions, Remote — Dec 2025 – Present**

- Build and maintain GenAI and agentic AI systems in LangChain, LangGraph, and Python across R&D initiatives and client-facing deliverables.
- Design RAG pipelines and **MCP server integrations** connecting enterprise tools to language models for production-ready workflows.
- Scope, prototype, and deploy AI-powered data-analysis tools that drive measurable business insights.

**GenAI Developer Contractor — Cognizant Technology Solutions, Remote — Jun 2025 – Dec 2025**

- Built and deployed custom **FastMCP servers** and external MCP integrations on AWS infrastructure to give agents dynamic tool access — modular, production-style agentic workflows.
- Delivered GenAI proof-of-concept solutions in LangChain/LangGraph/Python, accelerating team adoption of agentic AI patterns.
- Applied prompt engineering and model evaluation to optimize LLM outputs across business use cases.

## Projects

**CFO Agent — Personal Finance AI Platform (*Python, FastAPI, LangGraph, PostgreSQL, Redis, Anthropic API* — 2025–2026)**

- In active development: a personal-finance AI agent on FastAPI + LangGraph with PostgreSQL persistence and Redis caching, with CSV/OFX import and an encrypted manual data vault.
- Hybrid context architecture (auto aggregation + manual data “vault”) over sensitive financial data; architected for multi-tenant production.

**CreatorClip — AI Video Auto-Clipper / [autoclip.studio](#) (*Python, FastAPI, Celery, pgvector, WhisperX, Voyage AI, Anthropic API, ffmpeg, YouTube OAuth* — 2025–2026)**

- End-to-end production pipeline turning long-form YouTube content into short-form clips, learning a creator’s style via a custom “DNA” module.
- Async **Celery** pipeline: transcription (WhisperX), embeddings (Voyage + **pgvector**), rendering (ffmpeg), **YouTube OAuth** — a live SaaS (Stripe billing) in production.

**Agentic Software Development Team (*Python, LangChain, LangGraph, Amazon Bedrock* — *Cognizant R&D, 2025*)**

- Multi-agent LangGraph system with agents for code generation, testing, security review, and supervision — automating the full build lifecycle; Amazon Bedrock + OSS models as swappable inference backends.

## Technical Skills

- **AI / Agentic:** LangChain, LangGraph, RAG, MCP Servers (FastMCP), Amazon Bedrock, Anthropic API, Sentence Transformers, pgvector, prompt engineering, model evaluation
- **Integrations / Automation:** YouTube OAuth, MCP tool access, Celery async pipelines, WhisperX, Voyage AI, ffmpeg, CSV/OFX import
- **Backend / Data:** Python, FastAPI, Django REST, PostgreSQL, Redis, SQL · **Frontend:** React / TypeScript
- **DevOps / Cloud / ML:** Docker, AWS, Linux/Ubuntu, Git/GitHub, Azure Fundamentals, TensorFlow, Pandas, NumPy · **Tools:** VS Code, Claude Code, Jupyter

## Education

**BS Computer Science** — Shepherd University (GPA in major 3.43), Aug 2021 – May 2025

## Certifications

TensorFlow Developer Specialization (DeepLearning.ai) · Linux + Git (Boot.dev) · Azure Fundamentals (AZ-900) · GitHub Copilot · Generative AI Context Engineering (Cognizant)